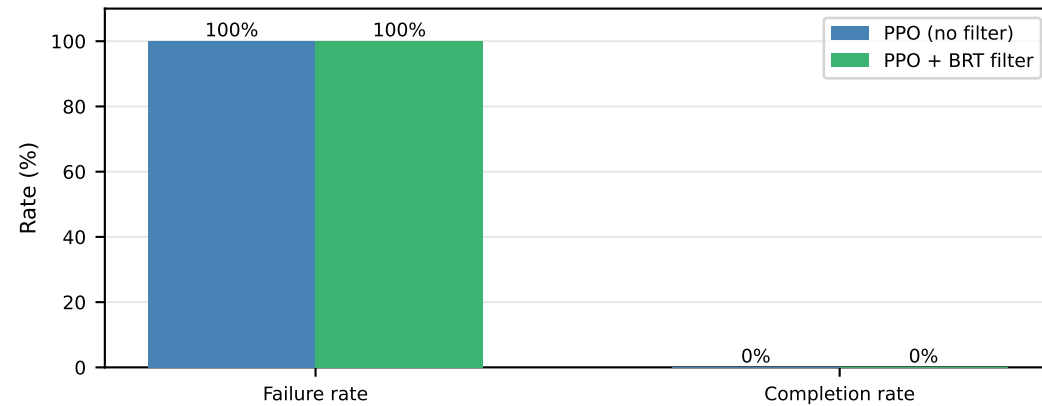
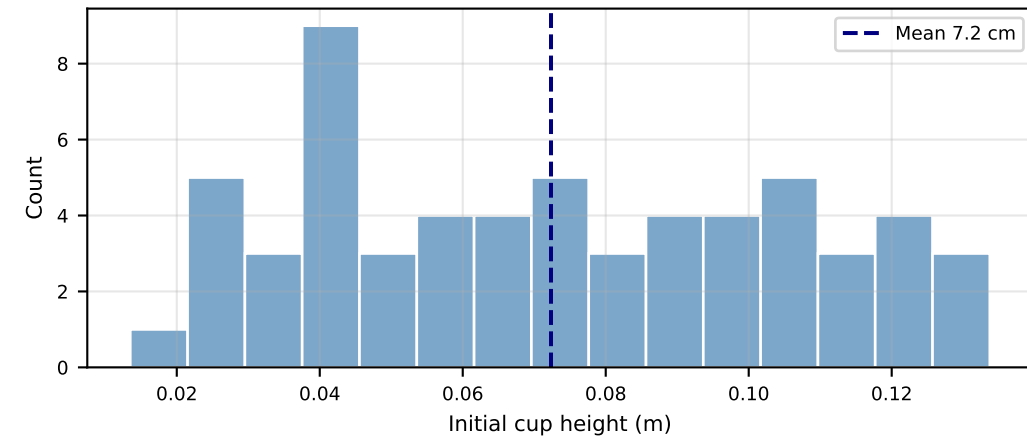


Hard-start evaluation ($e_l \in [2.0^\circ, 12.0^\circ]$, cup $z \approx 2\text{-}14\text{ cm}$ above ground)
PPO baseline: 100% failure | PPO + BRT: 100% failure (n=60 baseline, 60 filtered)

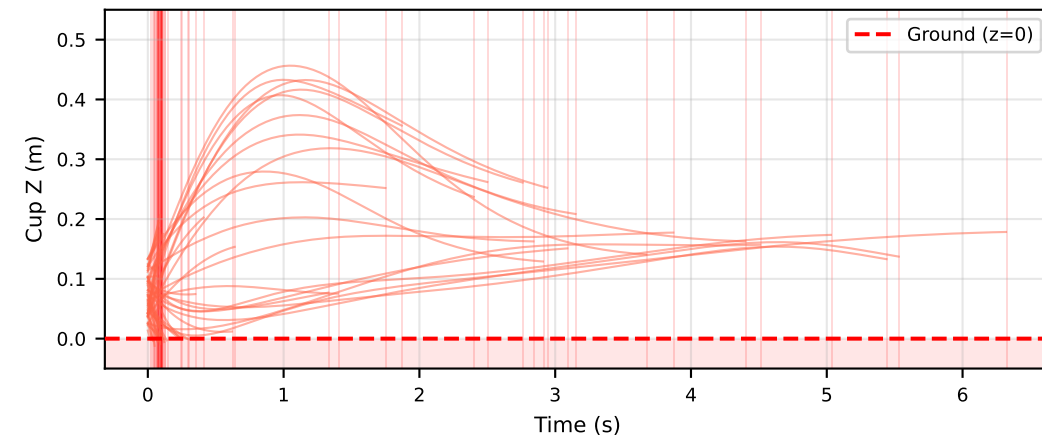
(A) Safety & task performance



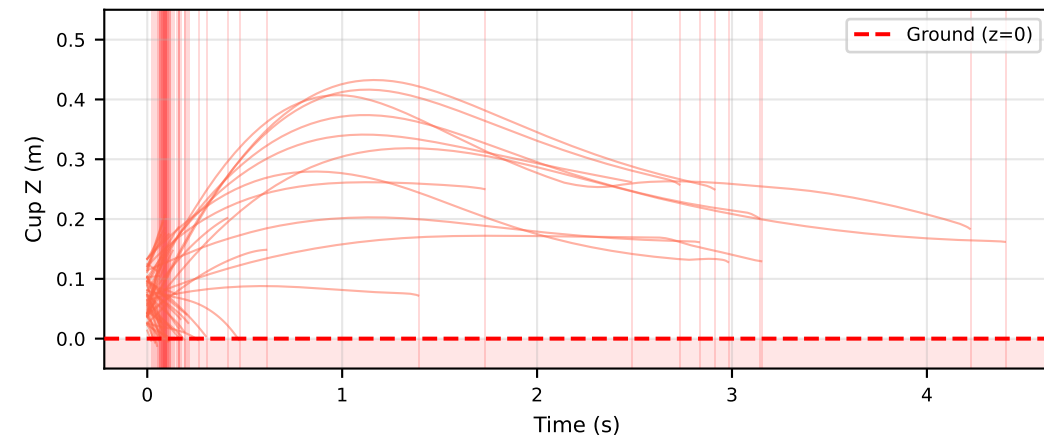
(B) Hard-start distribution



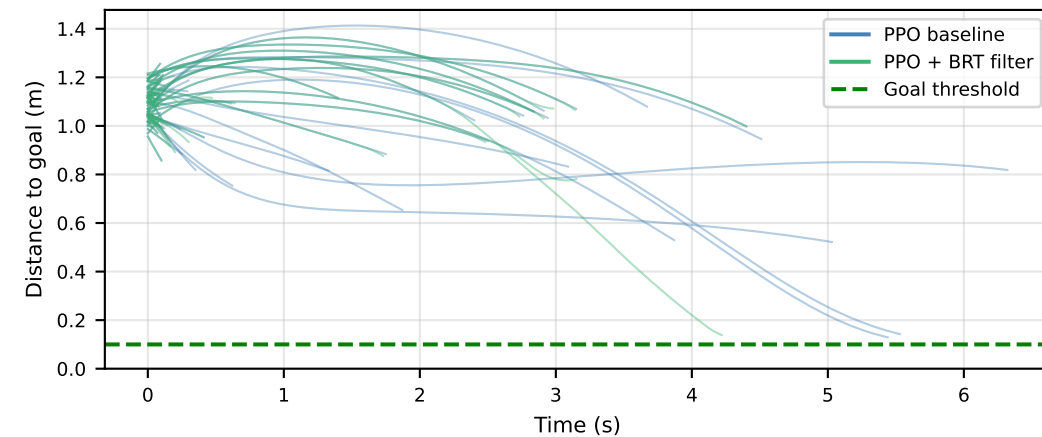
(C) PPO — cup height



(D) PPO + BRT filter — cup height



(E) Progress toward goal



(F) BRT filter activity per episode

